

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.49)

Metric

$$ds^2 = -dt \otimes dt + F(t)^2 dx \otimes dx + F(t)^2 H(x, y)^2 dy \otimes dy + V(t, x, y, z)^2 dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \right) dt + \left(\sqrt{\frac{1}{2}} F(t) \right) dx$$

$$n = \left(\sqrt{\frac{1}{2}} \right) dt + \left(-\sqrt{\frac{1}{2}} F(t) \right) dx$$

$$m = \left(\sqrt{\frac{1}{2}} F(t) H(x, y) \right) dy + \left(i \sqrt{\frac{1}{2}} \sqrt{V(t, x, y, z)^2} \right) dz$$

$$\bar{m} = \left(\sqrt{\frac{1}{2}} F(t) H(x, y) \right) dy + \left(-i \sqrt{\frac{1}{2}} \sqrt{V(t, x, y, z)^2} \right) dz$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2}}{2 F(t)} \right) dx$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{\sqrt{2}}{2 F(t)} \right) dx$$

$$m = \left(\frac{\sqrt{2}}{2 F(t) H(x, y)} \right) dy + \left(\frac{i \sqrt{2}}{2 |V(t, x, y, z)|} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2 F(t) H(x, y)} \right) dy + \left(-\frac{i \sqrt{2}}{2 |V(t, x, y, z)|} \right) dz$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2}}{2 F(t)} \partial_x$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t - \frac{\sqrt{2}}{2 F(t)} \partial_x$$

$$\delta = \frac{\sqrt{2}}{2 F(t) H(x, y)} \partial_y + \frac{i \sqrt{2}}{2 |V(t, x, y, z)|} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 F(t) H(x, y)} \partial_y - \frac{i \sqrt{2}}{2 |V(t, x, y, z)|} \partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)} + \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, x, y, z)}{4 V(t, x, y, z)} - \frac{\sqrt{2} \frac{\partial}{\partial x} H(x, y)}{4 F(t) H(x, y)} - \frac{\sqrt{2} \frac{\partial}{\partial x} V(t, x, y, z)}{4 F(t) V(t, x, y, z)}$$

$$\sigma = \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)} - \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, x, y, z)}{4 V(t, x, y, z)} - \frac{\sqrt{2} \frac{\partial}{\partial x} H(x, y)}{4 F(t) H(x, y)} + \frac{\sqrt{2} \frac{\partial}{\partial x} V(t, x, y, z)}{4 F(t) V(t, x, y, z)}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,x,y,z)}{4V(t,x,y,z)} - \frac{\sqrt{2}\frac{\partial}{\partial x}H(x,y)}{4F(t)H(x,y)} - \frac{\sqrt{2}\frac{\partial}{\partial x}V(t,x,y,z)}{4F(t)V(t,x,y,z)}$$

$$\lambda = -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)} + \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,x,y,z)}{4V(t,x,y,z)} - \frac{\sqrt{2}\frac{\partial}{\partial x}H(x,y)}{4F(t)H(x,y)} + \frac{\sqrt{2}\frac{\partial}{\partial x}V(t,x,y,z)}{4F(t)V(t,x,y,z)}$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\frac{\partial}{\partial y}V(t,x,y,z)}{4F(t)H(x,y)V(t,x,y,z)}$$

$$\beta = \frac{\sqrt{2}\frac{\partial}{\partial y}V(t,x,y,z)}{4F(t)H(x,y)V(t,x,y,z)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)}$$

Ricci Tensor Components

$$\begin{aligned}\Lambda = & \frac{\frac{\partial}{\partial t} F(t)^2}{12 F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{6 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, x, y, z)}{6 F(t) V(t, x, y, z)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, x, y, z)}{12 V(t, x, y, z)} - \frac{\frac{\partial^2}{(\partial x)^2} H(x, y)}{12 F(t)^2 H(x, y)} - \frac{\frac{\partial}{\partial x} H(x, y) \frac{\partial}{\partial x} V(t, x, y, z)}{12 F(t)^2 H(x, y) V(t, x, y, z)} \\ & - \frac{\frac{\partial^2}{(\partial x)^2} V(t, x, y, z)}{12 F(t)^2 V(t, x, y, z)} + \frac{\frac{\partial}{\partial y} H(x, y) \frac{\partial}{\partial y} V(t, x, y, z)}{12 F(t)^2 H(x, y)^3 V(t, x, y, z)} - \frac{\frac{\partial^2}{(\partial y)^2} V(t, x, y, z)}{12 F(t)^2 H(x, y)^2 V(t, x, y, z)}\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, x, y, z)}{4 F(t) V(t, x, y, z)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, x, y, z)}{4 V(t, x, y, z)} - \frac{\frac{\partial^2}{\partial t \partial x} V(t, x, y, z)}{2 F(t) V(t, x, y, z)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial x} V(t, x, y, z)}{2 F(t)^2 V(t, x, y, z)} - \frac{\frac{\partial^2}{(\partial x)^2} H(x, y)}{4 F(t)^2 H(x, y)} + \frac{\frac{\partial^2}{(\partial x)^2} V(t, x, y, z)}{4 F(t)^2 V(t, x, y, z)}$$

$$\Psi_1 = -\frac{\frac{\partial^2}{\partial t \partial y} V(t, x, y, z)}{4 F(t) H(x, y) V(t, x, y, z)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial y} V(t, x, y, z)}{4 F(t)^2 H(x, y) V(t, x, y, z)} + \frac{\frac{\partial^2}{\partial x \partial y} V(t, x, y, z)}{4 F(t)^2 H(x, y) V(t, x, y, z)} - \frac{\frac{\partial}{\partial x} H(x, y) \frac{\partial}{\partial y} V(t, x, y, z)}{4 F(t)^2 H(x, y)^2 V(t, x, y, z)}$$

$$\begin{aligned}\Psi_2 = & \frac{\frac{\partial}{\partial t} F(t)^2}{12 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{12 F(t)} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, x, y, z)}{12 F(t) V(t, x, y, z)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, x, y, z)}{12 V(t, x, y, z)} - \frac{\frac{\partial^2}{(\partial x)^2} H(x, y)}{12 F(t)^2 H(x, y)} + \frac{\frac{\partial}{\partial x} H(x, y) \frac{\partial}{\partial x} V(t, x, y, z)}{6 F(t)^2 H(x, y) V(t, x, y, z)} \\ & - \frac{\frac{\partial^2}{(\partial x)^2} V(t, x, y, z)}{12 F(t)^2 V(t, x, y, z)} - \frac{\frac{\partial}{\partial y} H(x, y) \frac{\partial}{\partial y} V(t, x, y, z)}{6 F(t)^2 H(x, y)^3 V(t, x, y, z)} + \frac{\frac{\partial^2}{(\partial y)^2} V(t, x, y, z)}{6 F(t)^2 H(x, y)^2 V(t, x, y, z)}\end{aligned}$$

$$\Psi_3 = -\frac{\frac{\partial^2}{\partial t \partial y} V(t, x, y, z)}{4 F(t) H(x, y) V(t, x, y, z)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial y} V(t, x, y, z)}{4 F(t)^2 H(x, y) V(t, x, y, z)} - \frac{\frac{\partial^2}{\partial x \partial y} V(t, x, y, z)}{4 F(t)^2 H(x, y) V(t, x, y, z)} + \frac{\frac{\partial}{\partial x} H(x, y) \frac{\partial}{\partial y} V(t, x, y, z)}{4 F(t)^2 H(x, y)^2 V(t, x, y, z)}$$

$$\Psi_4 = \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, x, y, z)}{4 F(t) V(t, x, y, z)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, x, y, z)}{4 V(t, x, y, z)} + \frac{\frac{\partial^2}{\partial t \partial x} V(t, x, y, z)}{2 F(t) V(t, x, y, z)} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial x} V(t, x, y, z)}{2 F(t)^2 V(t, x, y, z)} - \frac{\frac{\partial^2}{(\partial x)^2} H(x, y)}{4 F(t)^2 H(x, y)} + \frac{\frac{\partial^2}{(\partial x)^2} V(t, x, y, z)}{4 F(t)^2 V(t, x, y, z)}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"